

Dual-Masked and Discriminative Reconstruction for Unified Vision Anomaly Detection

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Abstract—Unsupervised reconstruction networks have shown promise for unified vision anomaly detection, *i.e.*, image-level anomaly classification and pixel-level anomaly segmentation, where a single model trained on multi-class normal images can detect various anomalies. This is more challenging than most existing separate methods, *i.e.*, one model for one class, as it requires handling a more complex data distribution. Notably, pure reconstruction networks often suffer from overfitting due to “identity shortcut”, where both normal and anomaly images may be well recovered and thus fail in detecting anomalies. Recent efforts have focused on developing specific modules for different network architectures, *e.g.*, Convolutions and Transformers. However, it is still unclear how to essentially and effectively prevent learning from this shortcut in a simpler and more general manner. Furthermore, most existing methods consider anomaly detection solely as unsupervised classification, resulting in inaccurate anomaly segmentation due to “weak discrimination”, where normal and anomaly features may be entangled. To address these challenges, we propose a simple yet general Dual-masked and Discriminative Reconstruction (D2Rec) for unified vision anomaly detection. First, we propose a general dual-masked reconstruction, *i.e.*, using a pair of complementary masks, resolving the “identity shortcut” so that all masked positions are reconstructed by unmasked original features. Second, we propose a self-supervised discriminator, which refines reconstruction errors with synthesized anomaly images to enhance the discrimination ability between normal and abnormal features. The dual-masked reconstruction and self-supervised discriminator can serve as universal plugins, easily integrated into reconstruction-based anomaly detection methods of any architecture. Despite its simplicity, D2Rec outperforms previous methods on three industrial benchmarks (MVTec, BTAD, and VisA), and three medical datasets (Brain MRI, Liver CT and Retinal OCT). The code for D2Rec is available at <https://github.com/gaobb/D2Rec>

Index Terms—Dual-masked reconstruction, self-supervised discriminator, unsupervised anomaly detection, unified anomaly detection.

I. INTRODUCTION

UNSUPERVISED vision anomaly detection aims to identify anomaly images (*i.e.*, classification or detection) and segment anomaly regions (*i.e.*, segmentation or localization) using only normal training images. It is increasingly emerging with wide applications, such as quality inspection in intelligent manufacturing [1], [2], [3], medical image diagnosis [4], [5], and video surveillance [6], [7], [8], [9], thus attracting more

research attention. Notably, most existing anomaly detection methods [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] mainly focus on training separate models (*i.e.*, one model for one class) for different objects. However, this separate paradigm may not be practical in real-world applications, as it requires high memory consumption and storage overhead, especially when increasing the number of categories. In contrast, unified anomaly detection (*i.e.*, one model for multiple classes) detects anomalies for multi-class objects with a single model. Notably, the categorical information (*i.e.*, class label) is inaccessible at both the training and inference stages in this unified setting. Undoubtedly, the unified paradigm is both more practical and more challenging, but it remains a relatively new and under-explored area.

UniAD [21] is the first unified unsupervised anomaly detection work, which is typically based on an encoder-decoder transformer. Notably, the pure encoder-decoder transformer suffers from overfitting. It cannot achieve satisfactory performance because of “identity shortcut” issue, which returns a direct copy of the input disregarding its content, as shown in Fig. 1a. Layer-wise query decoder and neighbor-masked attention are proposed in UniAD [21] to prevent the model from learning the shortcut. Similar to UniAD, DUMA [22] applies a max-mask matrix in self-attention to reduce the noise dependencies in transformer-based reconstruction networks, and SSPCAB [23] proposes a masked convolutional kernel for convolutional-based reconstruction networks. However, these methods are limited to specific network architectures, such as transformers or convolutions. Furthermore, RD [16] proposes a simple student-teacher reconstruction model, where the student restores the multiscale representations of the teacher from high-level to low-level. EfficientAD [19] combines a lightweight student-teacher model and an autoencoder model for efficient and accurate detection of vision anomalies. However, RD [16] and EfficientAD [19] are originally designed for separate settings, and there is still a large performance gap when they are extended to a unified setting. Here, we raise a question: *how to prevent the reconstruction networks regardless of architecture under a unified setting from falling into overfitting in a simpler and more general manner?*

Furthermore, the current unified anomaly detection methods, *e.g.*, UniAD [21] and MoEAD [24], are still not satisfactory for pixel-level anomaly segmentation when using a more suitable evaluation metric (see more detailed discussions in Sec. V), *e.g.*, Pixel-level AUPR, as shown in Figs. 1c and 2. The main reason is “weak discrimination” between normal and anomaly features because they are trained only on normal training images in low-resolution feature space. The “weak discrimination” makes normal and anomaly

Received 25 January 2025; revised 27 October 2025; accepted 5 April 2026. Date of publication 1 May 2026; date of current version 6 May 2026. The associate editor coordinating the review of this article and approving it for publication was Prof. Vittoria Bruni.

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TIP.2026.3687095>, provided by the authors.

Digital Object Identifier 10.1109/TIP.2026.3687095

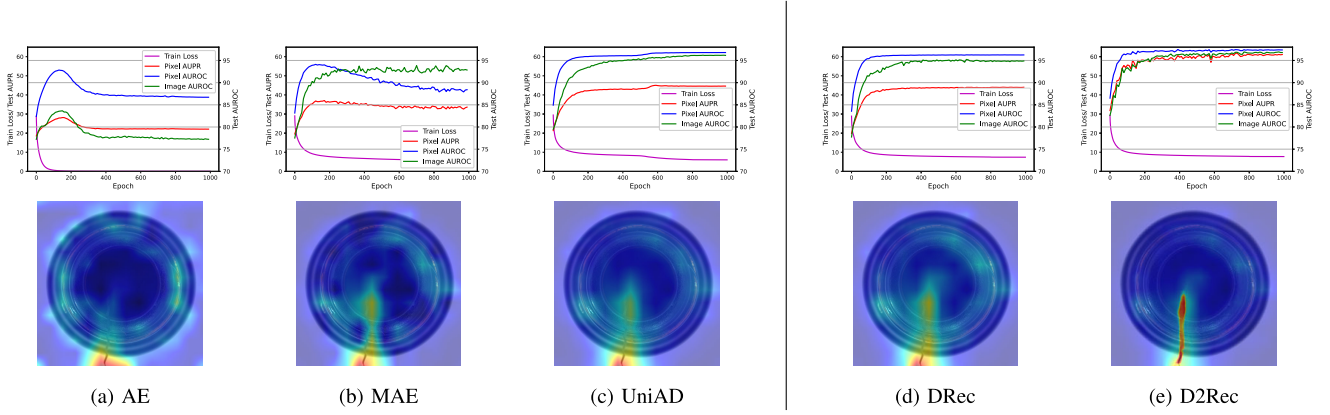


Fig. 1. **Comparisons among (a) AE, (b) MAE, (c) UniAD and our (d) DRec and (e) D2Rec with Eb4 +Trans architecture on MVTec.** The simple AE tends to learn an “identity shortcut”, *i.e.*, regressing the overall network to an identity mapping, which can result in the model copying input as output at inference, thereby failing to detect anomalies. We also observe that AE suffers from serious overfitting, *i.e.*, low training loss but poor testing performance, due to the “identity shortcut”. The UniAD can alleviate the “identity shortcut”, but it is primarily designed for transformer architectures. To some extent, Masked AE (MAE) can also mitigate the “identity shortcut”, but its effectiveness depends on an ideal mask ratio. In contrast, the proposed **Dual-masked Reconstruction (DRec)** is simple, general, and robust across various mask ratios. DRec effectively resolves the “identity shortcut” by reconstructing all masked positions using unmasked features. However, both UniAD and DRec still suffer from the “weak discrimination” issue, which leads to entanglement between normal and anomaly features, ultimately resulting in inaccurate anomaly segmentation. To this end, our D2Rec integrates **DRec** and **Self-Supervised Discriminator (SSD)** into a unified framework. This approach successfully resolves both the “identity shortcut” and “weak discrimination” issues simultaneously, achieving superior anomaly detection performance, particularly in anomaly segmentation.

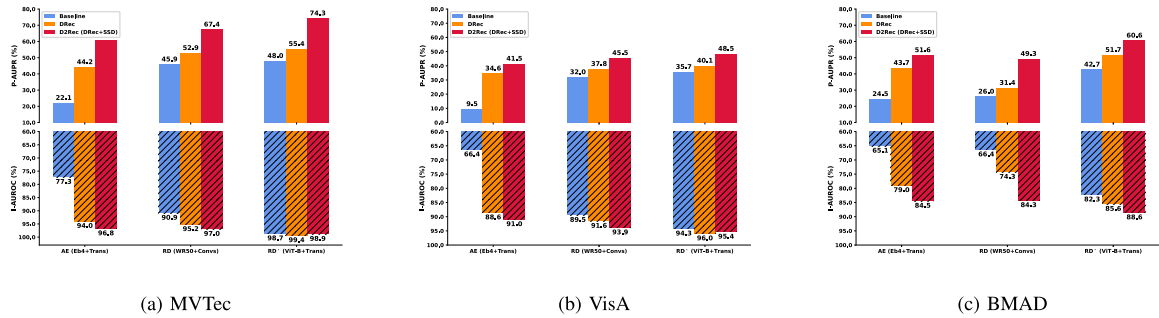


Fig. 2. **Comparisons of popular reconstruction-based anomaly detection models, e.g., AE and RD, and our DRec and D2Rec with diverse architectures.** The averaged results of all classes on each dataset are reported in **P-AUPR** for anomaly segmentation and **I-AUROC** for anomaly classification, respectively.

features always entangled, and ultimately results in inaccurate anomaly segmentation. We expect that anomaly segmentation performance can be improved if the “weak discrimination” issue is mitigated through discriminative learning. Recently, some discriminative models have been trained on synthetic anomalies generated on anomaly-free images [15], [20], [25], [26] or features [17]. However, current approaches either use synthetic anomalies at the image level, resulting in weak performance, or utilize a complex segmentation model, incurring high computational costs. *We expect to design a discriminator that is as lightweight as possible to refine reconstruction errors for achieving better anomaly segmentation.*

To overcome “identity shortcut” and “weak discrimination” issues for unified visual anomaly detection, we propose an embarrassingly simple but general D2Rec framework. First, we proposed a Dual-masked Reconstruction (DRec) mechanism, which ensures that all masked positions are reconstructed using unmasked features, thus completely solving the “identity shortcut” in unsupervised reconstruction. Experiments have demonstrated that the proposed DRec is robust and insensitive to mask ratio. Second, to further boost the anomaly segmentation performance caused by “weak discrimination” between normal and abnormal features, we

proposed a lightweight Self-Supervised Discriminator (SSD) to further refine the reconstruction error from low-resolution to high-resolution (coarse to fine) with normal and anomaly-free pseudo images. Our contributions are summarized as follows:

- We rethink and analyze reconstruction networks for unified anomaly detection, and then find that they severely suffer from overfitting and result in unsatisfactory performance due to “identity shortcut” and “weak discrimination”.
- We propose a simple and general dual-masked reconstruction for unsupervised reconstruction networks, which provides a unified solution to address the “identity shortcut” in reconstruction networks of any architecture.
- To alleviate “weak discrimination”, we propose a self-supervised discriminator to refine reconstruction errors with synthesized anomaly images, which significantly boosts anomaly segmentation performance.
- We propose a simple, effective, general and robust vision anomaly detection framework integrating unsupervised dual-masked reconstruction and self-supervised discriminator, and achieve competitive performance on both industrial and medical anomaly detection benchmarks.

II. RELATED WORK

Unsupervised vision anomaly detection methods can be mainly grouped into three types, *i.e.*, embedding-, discrimination-, and reconstruction-based methods.

A. Embedding-Based Methods

aim at leveraging deep models pre-trained on large-scale data to extract offline features from images for anomaly detection. It assumes that offline features preserve the discriminative information that helps distinguish anomalies from normal samples. PaDiM [11], MDND [27], and DFM [12] model a multivariate Gaussian distribution for normal features, then utilize a distance metric to measure anomalies. PatchCore [13] captures normal features and then stores them in a memory bank, and finally calculates anomaly scores by Euclidean distance between query features and the memory bank at inference. CS-Flow [10] and PyramidFlow [28] transform normal feature distribution into a Gaussian distribution via normalizing flow. CFA [14] and PADA [29] propose feature adaptation for adapting target datasets. Recent studies have shown that multiple designed or learnable text prompts on a powerful vision-language model [30], [31], [32] or only one normal image prompt on a pure vision model [33] can yield excellent performance for zero- and few-shot anomaly detection. However, these methods mainly focus on a separate setting, which has significant performance degradation when they are naively extended to a unified paradigm. While zero-/few-shot variants offer practical flexibility, they still lag behind full-shot methods. We focus on a unified setting and aim to achieve competitive performance with separate approaches.

B. Discrimination-Based Methods

are typically trained to distinguish normal images from synthesized anomaly images and hope that they can generalize to unseen real anomalies at test time. CutPaste [25] proposes a simple strategy to generate synthetic anomalies. It cuts a small rectangular area of variable sizes and aspect ratios from normal training images and pastes this patch back to the image at a random location. Similar to CutPaste, DRAEM [15] generates noise images using Perlin. A boundary-guided semi-push-pull loss is proposed to learn more discriminative features with synthetic samples in [34]. PRN [35] presents a variety of anomaly generation strategies for more accurate anomaly localization. Instead of synthesizing anomalies on images, SimpleNet [17] synthesizes anomaly features by adding Gaussian noise to normal features and then learns a binary discriminator to distinguish anomaly features from normal ones. Recently, GLASS [20] synthesizes global and local anomalies and achieves excellent performance in industrial anomaly detection and localization tasks. AnoGen [36] guides a diffusion model to generate realistic and diverse anomalies with only a few real anomaly images. Different from generating anomalies, this paper mainly aims to propose a simple, general, and lightweight self-supervised discriminator trained with normal and anomaly-free pseudo images to alleviate the issue of inaccurate anomaly segmentation.

C. Reconstruction-Based Methods

assume that anomaly image regions cannot be properly reconstructed since they do not exist in normal training images. Some works use generative models such as auto-encoders [37], [38], [39] and GANs [40], [41], [42] to reconstruct normal images. Recent works frame anomaly detection as an inpainting problem, where patches from images are partly masked. RIAD [43] randomly removes partial image regions and reconstructs the image from partial inpaintings with a convolutional neural network. SSPCAB [23] learns to reconstruct masked regions using contextual information with a masked convolutional kernel. RGI [44] proposes a robust GAN-inversion that can restore any input image (even with gross corruptions) to a clean image and identify the corrupted region mask by solving the optimization problems thereof. A pyramid deformation module is proposed to model diverse normal and measure the severity of anomaly in [45]. These methods tend to be computationally expensive because they involve reconstruction in image space.

UniAD [21] reconstructs features and achieves strong performance for unified anomaly detection. Moreover, RD [16] and EfficientAD [19] propose a simple and lightweight student-teacher reconstruction model, where the student restores the teacher's multiscale representations from high-level to low-level. Recently, some works, *e.g.*, DiAD [46] and OneNIP [47], try to use additional conditions or complex modules, such as normal image prompts and stable diffusion, to improve reconstruction models and achieve good performance for unified anomaly detection. However, these works are either designed for specific architectures, such as Trans or Convs, or are originally designed for separate settings. Therefore, they do not work well when extended to a unified setting or are difficult to generalize to different network architectures. This paper aims to propose a simple and general method for reconstruction networks of any architecture.

III. METHOD

The proposed Dual-masked and Discriminative Reconstruction (D2Rec) jointly learns by reconstruction and discrimination for unified vision anomaly detection, as shown in Fig. 3. It is composed of a multi-level features fusion, a dual-masked reconstruction, and a self-supervised discriminator. We elaborately introduce them in this section.

A. Multi-Level Features Fusion

Following existing anomaly detection works, we extract image features using pre-trained models. Given an input image $I \in \mathbb{R}^{3 \times H \times W}$, we extract multi-level features $\{Z_i\}$ from multiple stages of a pre-trained backbone $\mathcal{F}$, that is

$$Z_i = \mathcal{F}(I), \quad (1)$$

where $Z_i \in \mathbb{R}^{c_i \times h_i \times w_i}$, c_i is the channel number, and $h_i \times w_i$ is the spatial shape of the i -th level feature. Considering that multi-level features $\{Z_i\}_{i=1}^L$ usually represent hierarchical semantic information and their shapes may be different, we

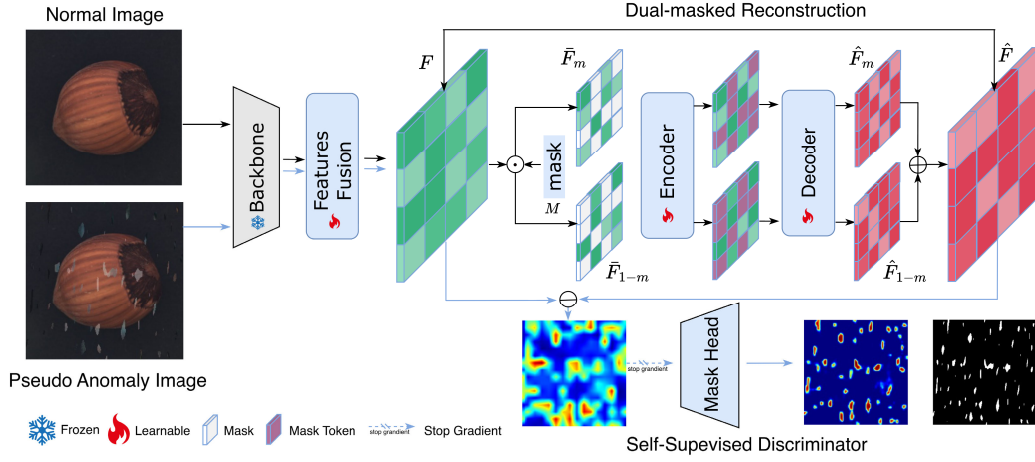


Fig. 3. **Framework of the proposed D2Rec for unified vision anomaly detection.** In the training stage, both normal and synthetic pseudo anomaly images are fed a pre-trained backbone for extracting multi-level features. These features are transformed into a unified space to obtain compact representations with multi-level features fusion. Next, a dual-masked reconstruction is utilized to reconstruct normal features, and a self-supervised discriminator with a mask head is used to refine the reconstruction errors from coarse to fine. At inference, the synthesized images are removed, and only testing images are fed into D2Rec. The final prediction is a weighted combination of the reconstruction error and the segmentation map from the mask head.

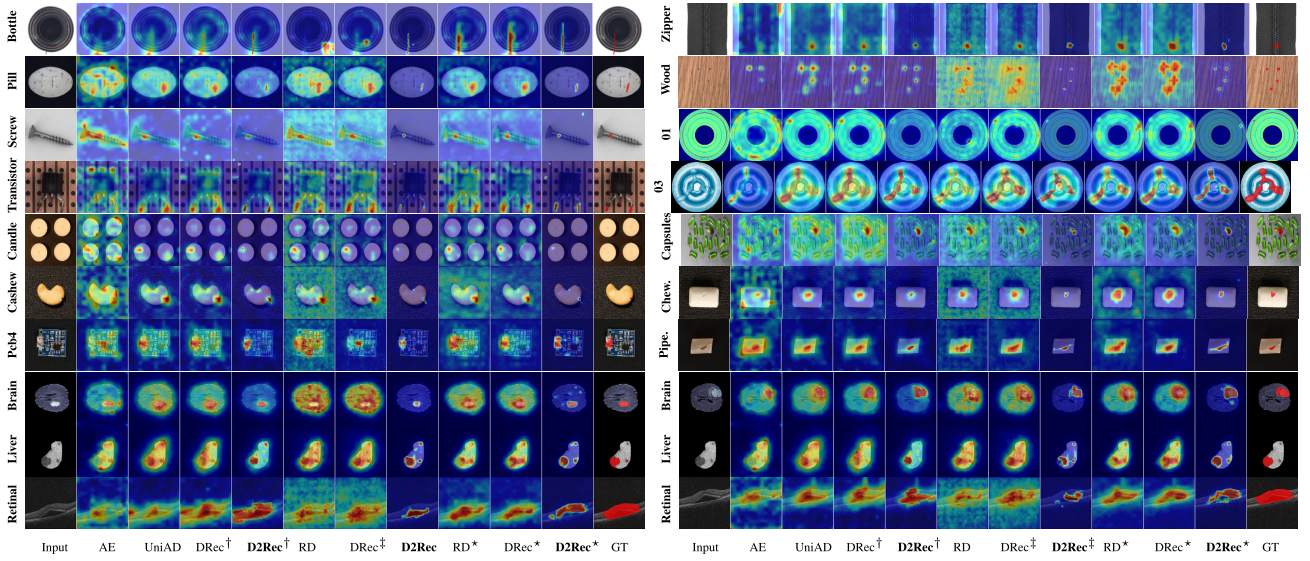


Fig. 4. Visualization comparisons of several baselines and our DRec and D2Rec with diverse network architectures on selected testing images from MVTEC [1], BTAD [2], VisA [3] and BMAD [50] datasets. Here, the superscript [†], [‡] and ^{*} means Eb4 +Trans, WR50 +Convs and ViT-B+Trans, respectively.

have to fuse them into a unified space to obtain a compact representation F , that is

$$F = \text{Fuse}\{Z_i\}_{i=1}^l. \quad (2)$$

For a fair comparison, we adopt the same multi-scale feature fusion strategy consistent with the state-of-the-art methods, which varies across backbones and model architectures. The respective fusion details are described in Section IV-A.

B. Dual-Masked Reconstruction

MAE reconstructs the original signal given its partial observation. It has proven to be a powerful self-supervised method for both computer vision [48] and natural language processing. Similar to all autoencoders, MAE has an encoder-decoder architecture, where the encoder maps the observed signal to a latent representation and the decoder reconstructs the original signal from the latent representation. The difference

is that MAE adopts an asymmetric encoder-decoder, which allows the encoder to operate only on a small portion of the observed signal through random masking with a mask ratio, and the decoder reconstructs the full signal from the latent representation and learnable mask tokens. We empirically found that pure MAE is only effective when using a high mask ratio, but its performance is highly sensitive to mask ratios and still suffers from overfitting, as shown in Fig. 6.

In MAE, unmasked features are easily reconstructed by learning a shortcut. To avoid overfitting (shortcut), we propose a Dual-masked Reconstruction (DRec), which works well with various mask ratios for unsupervised vision anomaly detection. The DRec consists of a dual-masked encoder/decoder.

1) *Dual-Masked Encoder*: The dual-masked encoder is applied only to visible (*i.e.*, unmasked) features. The encoder embeds the compact representation F derived from multi-level features fusion. Different from the general encoder, our

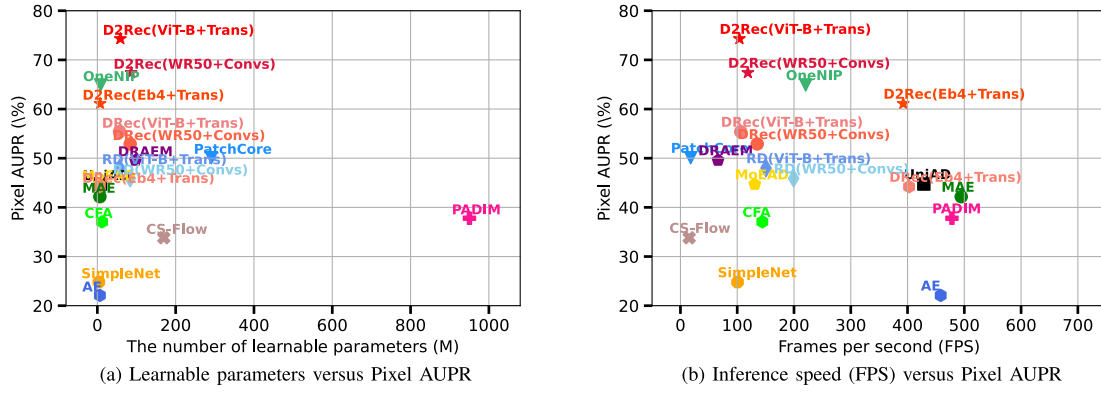


Fig. 5. Comparisons of Performance, Complexity and Efficiency of different methods with a unified setting on MVTec [1]. Note that M means million (10^6) and FPS means the number of inferencing images in one second on one NVIDIA V100 GPU, with a batch size of 64. Our methods (DRec and D2Rec) equipped with a lightweight network architecture (Eb4 +Trans) outperform most previous ones in Pixel AUPR (about 10%) with comparable parameters (7.1M) and inference speed (400 + FPS). The best anomaly detection performance is obtained by our D2Rec when using a more powerful network (ViT-B+Trans) but with acceptable parameter size (58.8M) and inference speed (104 FPS).

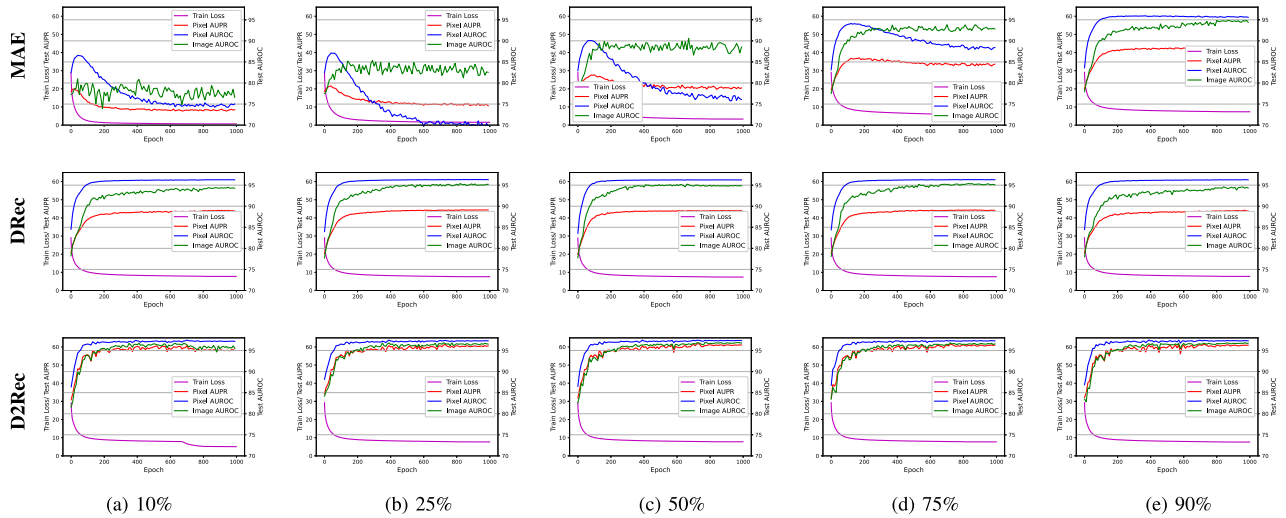


Fig. 6. **Comparisons of training loss and testing metrics** on MVTec using MAE, DRec, and D2Rec, with varying mask ratios and Eb4 +Trans architecture. Notably, dual mask ratios are used in D2Rec, *e.g.*, 10% corresponds to 10% and 90%. The MAE only works well with a high masking ratio, *e.g.*, 90%, as shown in the last of the first row, while lower mask ratios still suffer from overfitting. In contrast, DRec demonstrates robust performance across various mask ratios, suggesting that the dual-masked mechanism effectively addresses the over-fitting issue of MAE. Moreover, D2Rec shows significant improvements in pixel-level AUPR across all mask ratios, indicating that the self-supervised discriminator enhances the discriminability between normal and anomaly features. Consequently, D2Rec is most effective when integrating dual-masked reconstruction and self-supervised discriminator.

encoder only operates on a partial subset (*e.g.*, 50%) of the full features, *i.e.*, any spatial features are in F , through random masking $M \in \mathcal{R}^{h \times w}$, where each element of M is either 0 or 1. Therefore, the input of our encoder is

$$\bar{F}_m = M \odot F, \quad (3)$$

where $\odot$ represents an element-wise multiplication and masked tokens are removed from $\bar{F}_m$. For simplicity, the corresponding latent representation of the encoder is still denoted as $\bar{F}_m$. Notably, the masked encoder operates on a subset of the visible features and reconstructs the full features, including both visible (unmasked) and invisible (masked) features, with another decoder described below. To some extent, the random masking operation alleviates over-fitting during the reconstruction process. *However, the issue of over-fitting still exists because the observed features can be completely reconstructed by simply copying inputs as the outputs, especially for low mask ratios.* To alleviate this issue, we propose a dual mask

as a complementary component of common random masking as follows:

$$\bar{F}_{1-m} = (1 - M) \odot F. \quad (4)$$

Combining Eqs. 3 and 4 together (*i.e.*, dual-masked encoder), we can ensure that all masked (reconstructed) features depend on unmasked ones rather than themselves. Therefore, the overfitting issue in MAE may be thoroughly resolved by using the proposed dual-masked mechanism.

2) *Dual-Masked Decoder*: The input to our dual-masked decoder is a full set consisting of latent representations $\bar{F}_m$ derived from the dual-masked encoder on visible features and mask tokens $T \in \mathcal{R}^c$. Each mask token T is a shared and learned embedding that indicates the presence of invisible features to be predicted. In our implementation, we first restore the latent representation $\bar{F}_m$ into the original spatial location depending on the mask M , and fill the remaining locations with this shared mask token. Notably, the mask token would have

no location information, and we add positional embeddings to the mask token in this constructed full set. Then, we feed the resulting full set via the decoder. Here, we denote the output of the masked-decoder as $\hat{F}_m^{c \times h \times w}$. On the other hand, we can derive another reconstructed feature $\hat{F}_{1-m}^{c \times h \times w}$ adopting the above decoder processing on another latent representation $\hat{F}_{1-m}$.

Notably, $\hat{F}_m$ and $\hat{F}_{1-m}$ are reconstructed by visible $\bar{F}_m$ and $\bar{F}_{1-m}$, respectively. In other words, the visible features could be completely reconstructed by copying the input as the output. In order to avoid the issue, we combine $\hat{F}_m$ and $\hat{F}_{1-m}$ with their corresponding opposite masks, that is

$$\hat{F} = (1 - M) \odot \hat{F}_m + M \odot \hat{F}_{1-m}. \quad (5)$$

3) *Reconstruction Function*: The reconstruction function is used to measure pixel-wise errors between the reconstructed output $\hat{F}$ of the dual-masked decoder and the original input features F . In Eq. 5, each element of $\hat{F}$ is reconstructed by its corresponding unmasked features without relying on its own original (*i.e.*, visible or masked) features. *Therefore, it thoroughly solves the “identity shortcut” because all masked positions are reconstructed by unmasked original features, where there is no overlap between masked and unmasked features.* We use the same loss function as state-of-the-arts to measure reconstruction errors between the reconstructed and original features as

$$\mathcal{L}_{rec}(\hat{F}, F), \quad (6)$$

where $\mathcal{L}_{rec}$ is either mean squared error or cosine similarity, depending on the used framework. The loss details for different frameworks are described in Section IV-A.

C. Self-Supervised Discriminator

The DRec partially solves “identify shortcut” and achieves good performance on both I- and P-AUROC for unified anomaly detection. However, it still performs poorly when using a more rigorous metric (*i.e.*, P-AUPR) as shown in Fig. 1d. It is not surprising that DRec (like UniAD) is only trained on normal training images, which may result in “weak discrimination” between normal and abnormal features.

To enhance the discrimination between normal and anomaly features, the easiest way is to learn a binary classifier by normal and anomaly training images. Unfortunately, there are only normal images for unsupervised anomaly detection tasks. This is, in fact, also a sound setting for practical applications, such as industry inspection and medical image diagnosis, because anomaly images are always rare and difficult to collect on a large scale. Therefore, we have to synthesize anomaly images using normal ones for learning a discriminator. Fortunately, some researchers have tried to synthesize anomaly images and achieved good performance in anomaly detection fields, such as CutPaste [25] and DRAEM [15]. In this paper, we directly utilize CutPaste [25] and DRAEM [15] to synthesize anomaly images. Different from CutPaste [25], we learn a pixel-level discriminator with these synthesized anomaly images. Furthermore, we design a lightweight discriminator to refine the dual-masked reconstruction errors, which is also different from DRAEM [15], using a heavy segmentation model in image space.

Given a normal training image I^n and the corresponding anomaly mask Y^n , we denote its synthesized anomaly image and anomaly mask as I^a and Y^a , respectively. We feed the synthesized anomaly image I^a into a pre-trained backbone and derive multi-level features $\{Z_i^a\}$. Then, we can derive the compact representation F^a applying Eq. 2. Then, we reconstruct F^a with the proposed DRec, and denote the corresponding features as $\hat{F}^a$. Here, we use an element-wise reconstruction error E^a between reconstructed and original features, that is

$$E^a = \text{Error}(\hat{F}^a, F^a). \quad (7)$$

To ensure stable training, we set the reconstruction error E^a matching the corresponding reconstruction loss, *i.e.*, Eq. 6.

The discriminator is designed with several convolution blocks following a 1×1 convolution layer for performing pixel segmentation. Here, each convolution block is composed of a 3×3 convolution, a BatchNorm, a ReLU, and a 2×2 deconvolution. The reconstruction error E^a is fed into the designed discriminator to obtain an estimated anomaly map as $\hat{Y}^a$. For computing the loss between $\hat{Y}^a$ and the ground-truth Y^a , we resize $\hat{Y}^a$ to the size of Y^a . Considering that anomaly pixels are typically in the minority in anomaly detection, we utilize Dice loss [49], which is effective for learning from extremely imbalanced data, that is

$$\mathcal{L}_{seg} = 1 - \frac{2 \cdot \hat{Y}^a \cdot Y^a}{(\hat{Y}^a)^2 + (Y^a)^2}, \quad (8)$$

D. Training

Considering both the reconstruction and discrimination objectives of the two sub-networks, the total loss used in training D2Rec is

$$\mathcal{L} = \mathcal{L}_{rec} + \lambda \mathcal{L}_{seg}, \quad (9)$$

where $\lambda > 0$ is a weight that balances the importance of the two loss functions $\mathcal{L}_{rec}$ and $\mathcal{L}_{seg}$.

E. Inference

1) *Anomaly Segmentation*: The result of anomaly segmentation is an anomaly score map, which assigns an anomaly score for each pixel. For the dual-masked reconstruction, the anomaly score map, S_{rec} , is calculated by the reconstruction error. For the self-supervised discriminator, the anomaly score map is predicted as the $\hat{Y} \in \mathbb{R}^{H \times W}$. Finally, we combine S_{rec} and $\hat{Y}$ and take it as the final anomaly segmentation map, that is

$$S = w \cdot S_{rec} + (1 - w) \cdot \hat{Y}, \quad (10)$$

where $w \in [0, 1]$ is a weight.

2) *Anomaly Classification*: Anomaly classification aims to detect whether an image contains anomaly regions. We transform the anomaly score map S to the anomaly score of the image by taking the maximum value of S .

IV. EXPERIMENTS

A. Experimental Setup

We follow the previous works and evaluate our method on three popular industry anomaly detection benchmarks,

MVTec [1], BTAD [2], and VisA [3], and three medical datasets [50], Brain MRI, Liver CT and Retina OCT.

Protocol: Following the unified anomaly detection protocol in UniAD, we train a single model for detecting all categories. For fair comparisons, we use the original training/testing splits given in previous works [1], [2], [3], [50].

Metric: We evaluate and compare state-of-the-art anomaly detection methods and our D2Rec for image-level anomaly classification and pixel-level anomaly segmentation with AUPR and AUROC metrics. It is worth noting that the AUPR metric is better for anomaly segmentation measurement due to the imbalance issue between normal and anomaly pixels [3], [51].

Competed Methods: Our approach is compared with popular anomaly detection methods including CS-Flow [10], PaDiM [11], DFM [12], PatchCore [13], CFA [14], DRAEM [15], SimpleNet [17], RD [16], UniAD [21], MoEAD [24] and OneNIP [47]. To fully verify the effectiveness of our method, we insert the proposed D2Rec into several popular baselines, AE [21], RD [16], and RD*, using diverse backbones and reconstruction networks, *e.g.*, EfficientNet-b4 + Transformers, WideRes-50 + Convolutions, ViT-B/14 + Transformers (denoted as Eb4 +Trans, WR50 +Convs and DINOv2 ViT-B+Trans for simplicity). The details of these baselines are summarized as follows:

UniAD [21] is the first work for unified anomaly detection. It builds on an encoder-decoder transformer architecture and an EfficientNet-b4 [52] backbone. To prevent the “identity shortcut”, UniAD introduces layer-wise query and neighborhood-masked attention. **AE** is a degraded version of UniAD. To ensure fair comparisons, we use the same hyperparameters and network architecture as in UniAD [21] but fully remove all components in UniAD. Following UniAD, we extract multi-scale features $\{Z_i\}_{i=1}^4$ from the 1-th to 4-th stage of a pre-trained EfficientNet-b4. All these features are resized to the same size ($h \times w$), *i.e.*, the size of the smallest one. Then, we concatenate these resized features in the channel dimension to obtain the fusion representation F . The reconstruction loss, Eq. 6, is set to the mean squared error (MSE), and the error, Eq. 7, is measured by the absolute element-wise subtraction, respectively.

RD [16] is convolution-based reconstruction network using WideRes-50 backbone. It is originally designed for a separate setting, and here we extend it to the unified paradigm. Instead of reconstructing a compact feature in AE and UniAD, RD takes the compact feature as input and reconstructs multi-level original features in a reverse manner. Following RD [16], we extract multi-scale feature maps $\{Z_i\}_{i=1}^3$ from the 1-th to 3-th stage of a pre-trained WideRes-50. Different from the simple resizing operation in UniAD, RD utilizes two trainable blocks, MFF and OCE, for obtaining the compact representation F . We use cosine similarity as reconstruction loss and error in Eqs. 6 and 7.

RD* is a simple extension of RD using a more powerful backbone, DINOv2 ViT-B/14 [53], and a transformer decoder. Unlike convolution-based EfficientNet-b4 and WideRes-50, DINOv2 ViT-B/14 adopts full transformer architectures, and all features from different stages share the same dimensions.

Here, the multi-level features are extracted from the intermediate 8 stages, *i.e.*, $\{2, 3, \dots, 9\}$. For simplicity, we average these features from different stages to obtain the final representation F . Furthermore, we built a reconstruction decoder with the standard ViT block, which is also different from the original convolution block. The cosine similarity is used to measure reconstruction loss and error in Eqs. 6 and 7, which is the same as RD [16].

Implementation Details: Unless otherwise specified, we resize input images to 224×224 resolution for all methods both training phase and inference time. For DRec and D2Rec, we set the layer numbers of the encoder and decoder to 4 for Eb4 +Trans, and build the encoder for the identity function for WR50 +Convs and ViT-B+Trans, respectively. For the self-supervised discriminator, the number and dimension of the convolution block are set to 2 and 128 for balancing performance and computation costs. The loss weight λ in Eq. 9 and coefficient w in Eq. 10 are set to 0.5 by default, respectively.

We conduct experiments based on the open-source framework PyTorch and NVIDIA V100 GPU. For Eb4 +Trans, we train the reconstruction model with a total of 1000 epochs on 8 Tesla V100 GPUs with batch size 64, and use AdamW optimizer with weight decay 1×10^{-4} . The learning rate is 1×10^{-4} initially and drops by 0.1 after 800 epochs. For WR50 +Convs and ViT-B+Trans, we train the reconstruction model with a total of 50 epochs on 1 Tesla V100 GPUs with batch size 16, and use AdamW optimizer with weight decay 1×10^{-4} . The initial learning rate is set to 2×10^{-3} and using a cosine annealing learning rate scheduler. We use the official codes for DRAEM [15], SimpleNet [17], RD [16], UniAD [21], MoEAD [24] and OneNIP [47], and publicly available implementations for other methods (*e.g.*, CS-Flow [54], PaDiM [11], PatchCore [13] and CFA [14]).

B. Comparisons With State-of-the-Arts

Results of industrial anomaly segmentation on MVTec, BTAD and VisA are reported in Tab. I. **1)** Most baseline and state-of-the-art methods with a unified setting (one model for multiple classes) suffer from a performance drop compared to the original results reported in their papers with one model for one class setting, which is also consistent with observations in UniAD. For example, state-of-the-art SimpleNet [17] drops about 17% (from 98.1% to 81.0%) in pixel-level AUROC. **2)** The pixel-level AUPR performance is far lower than AUROC for all methods under a unified setting (*e.g.*, less than 50%). This indicates that AUPR metric may be better than AUROC for measuring pixel-level anomaly segmentation. **3)** Our method beats all baselines and achieves the best pixel-level AUPR on two widely used industrial benchmarks, 74.3% on MVTec, and 48.5% on VisA. Our method significantly outperforms state-of-the-art by a larger margin. On BTAD, the anomaly segmentation performance of our D2Rec is slightly weaker than its baseline (RD) when using a reverse distillation network with a powerful ViT-B+Trans. We hypothesize that the limitations may be attributed to the annotation quality of the dataset, as BTAD contains a certain level of noise [55].

TABLE I

PIXEL-LEVEL ANOMALY SEGMENTATION (UPPER PART) AND IMAGE-LEVEL ANOMALY CLASSIFICATION (BOTTOM PART) COMPARISON WITH AUROC/AUPR ON THREE INDUSTRIAL BENCHMARKS, MVTEC, BTAD AND VISA. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Net Archs.	MVTEC		BTAD		VisA		Mean	
CS-Flow [10]	AlexNet+Convs	93.8	33.8	95.9	34.6	95.6	18.6	95.1	29.0
PaDiM [11]	WR50+Gaussian	95.5	37.8	96.7	48.7	95.9	17.1	96.0	34.5
DFM [12]	R50+Gaussian	96.5	42.4	96.3	48.0	96.5	25.2	96.4	38.5
PatchCore [13]	WR50+NN	96.4	50.1	96.3	48.4	96.8	38.2	96.5	45.6
CFA [14]	WR50+Convs	90.7	37.1	95.6	40.4	91.4	16.8	92.6	31.4
DRAEM [15]	Scratch+Convs	85.2	49.6	74.2	12.3	78.1	15.1	79.2	25.7
SimpleNet [17]	WR50+Linear	81.0	24.8	78.8	36.2	95.3	33.1	85.0	31.4
MoEAD [24]	Eb4+Trans	97.0	44.7	97.1	50.9	98.5	32.5	97.5	42.7
OneNIP [47]	Eb4+Trans	97.9	63.9	97.4	56.3	98.7	42.5	98.0	54.2
AE	Eb4+Trans	86.7	22.1	96.2	39.6	90.9	9.5	91.3	23.7
UniAD [21]	Eb4+Trans	96.8	44.7	97.1	51.9	98.4	33.6	97.4	43.4
D2Rec	Eb4+Trans	97.4	61.1	97.2	53.5	98.5	41.5	97.7	52.0
RD[16]	WR50+Convs	94.7	45.9	97.4	48.8	97.0	32.0	96.4	42.2
D2Rec	WR50+Convs	97.7	67.4	97.4	50.9	98.5	45.5	97.9	54.6
RD*	ViT-B+Trans	97.5	48.0	97.8	63.3	98.5	35.7	97.9	49.0
D2Rec	ViT-B+Trans	98.9	74.3	97.6	61.3	99.0	48.5	98.5	61.4
CS-Flow [10]	AlexNet+Convs	81.4	90.2	91.8	96.3	75.8	80.0	83.0	89.8
PaDiM [11]	WR50+Gaussian	87.5	92.8	95.7	97.4	78.1	78.3	87.1	92.4
DFM [12]	R50+Gaussian	69.7	89.8	68.7	82.8	51.6	77.8	63.3	78.6
PatchCore [13]	WR50+NN	89.8	96.3	95.9	97.4	90.3	92.0	92.0	95.2
CFA [14]	WR50+Convs	80.4	91.0	87.5	87.7	69.0	73.8	79.0	85.9
DRAEM [15]	Scratch+Convs	91.4	95.3	84.7	95.0	81.8	85.8	86.0	92.1
SimpleNet [17]	WR50+Linear	78.2	90.0	90.3	95.0	89.2	92.2	85.9	90.3
MoEAD [24]	Eb4+Trans	97.7	99.2	92.9	98.4	92.0	94.0	94.2	97.3
OneNIP [47]	Eb4+Trans	98.0	99.3	93.2	98.5	92.8	94.8	94.7	97.5
AE	Eb4+Trans	77.3	89.6	91.7	95.5	66.4	70.9	78.5	87.9
UniAD [21]	Eb4+Trans	96.7	98.9	92.5	97.9	90.8	93.0	93.3	96.7
D2Rec	Eb4+Trans	96.8	98.9	93.7	98.8	91.0	93.2	93.8	97.2
RD[16]	WR50+Convs	90.9	95.0	93.7	98.8	89.5	91.0	91.4	95.1
D2Rec	WR50+Convs	97.0	98.6	94.2	97.2	93.9	95.0	95.0	96.9
RD*	ViT-B+Trans	98.7	99.1	95.0	97.5	94.3	95.4	96.0	97.5
D2Rec	ViT-B+Trans	98.9	99.6	96.2	96.6	95.4	96.3	96.8	97.7

Results of industrial anomaly classification on MVTEC, BTAD and VisA are presented in Tab. I. 1) We observe a significant drop in anomaly classification performance when extending separated methods (one model for one class) to the unified setting (one model for multiple classes), which is consistent with the findings for anomaly segmentation. 2) Our method outperforms all baselines and state-of-the-arts and achieves the best average AUROC and AUPR scores on MVTEC, BTAD and VisA. It is noteworthy that our method only introduces a dual-masked mechanism to reconstruction networks and refines the reconstruction errors with self-supervised discriminator, supporting any network architectures, *e.g.*, convolution and transformer. Compared to UniAD [21], our approach is simpler and more general and can be considered as an alternative to UniAD for different network architectures.

Results of medical anomaly classification and segmentation on BMAD [50] are reported in Tab. II. First, the performance trend of our method on the medical dataset is consistent with that of industrial datasets. It significantly outperforms the corresponding baseline methods including AE, UniAD [21], RD [16] and RD* in both image-level anomaly classification and pixel-level anomaly segmentation. Second, our method can be flexibly scaled to stronger backbones, such as DINOv2 ViT-B/14, and more diverse reconstruction networks, such as transformer-based reversed distillation, and can steadily improve the performance of image-level anomaly classification (from 82.3% to 88.6% in AUROC) and pixel-level anomaly segmentation (from 42.7% to 60.6% in AUPR).

TABLE II

PIXEL-LEVEL ANOMALY SEGMENTATION (UPPER PART) AND IMAGE-LEVEL ANOMALY CLASSIFICATION (BOTTOM PART) COMPARISONS WITH AUROC/AUPR ON THREE MEDICAL DATASETS, INCLUDING BRAIN MRI, LIVER CT AND RETINAL OCT, FROM THE BMAD [50]. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Net Archs.	Brain		Liver		Retinal		Mean	
SimpleNet[17]	WR50+Linear	94.3	37.7	97.2	14.0	86.8	14.0	92.8	33.0
MoEAD[24]	Eb4+Trans	97.6	56.6	97.7	10.3	96.0	59.4	97.1	42.1
OneNIP[47]	Eb4+Trans	98.0	63.9	97.4	10.6	96.4	63.6	97.3	46.1
AE	Eb4+Trans	95.2	47.8	95.2	3.9	87.1	21.7	92.5	24.5
UniAD[21]	Eb4+Trans	97.7	59.7	97.1	7.1	95.8	54.5	96.9	40.4
D2Rec	Eb4+Trans	98.6	76.0	97.8	16.8	95.9	62.1	97.5	51.6
RD[16]	WR50+Convs	89.1	14.5	97.3	7.7	94.6	55.9	93.7	26.0
D2Rec	WR50+Convs	94.2	48.9	98.0	33.8	96.5	65.1	96.2	49.3
RD*	ViT-B+Trans	96.8	45.3	97.2	8.8	97.0	74.0	97.0	42.7
D2Rec	ViT-B+Trans	98.8	78.6	98.6	36.1	96.7	67.1	98.0	60.6
SimpleNet[17]	WR50+Linear	85.1	96.3	60.5	56.4	82.1	79.5	75.9	77.4
MoEAD[24]	Eb4+Trans	90.9	97.5	71.0	60.8	87.2	83.8	83.0	80.7
OneNIP[47]	Eb4+Trans	93.7	98.6	68.0	59.3	90.7	89.3	84.1	82.4
AE	Eb4+Trans	76.7	93.8	54.4	46.4	64.2	58.0	65.1	66.1
UniAD[21]	Eb4+Trans	91.6	98.1	63.7	50.5	86.2	83.6	80.5	77.4
D2Rec	Eb4+Trans	93.6	98.5	71.5	69.7	88.5	87.3	84.5	85.2
RD[16]	WR50+Convs	58.3	82.7	54.5	44.8	86.3	81.1	66.4	69.5
D2Rec	WR50+Convs	83.8	95.5	80.2	77.8	88.9	84.8	84.3	86.0
RD*	ViT-B+Trans	86.4	96.3	66.6	56.0	94.1	92.5	82.3	81.6
D2Rec	ViT-B+Trans	95.1	98.9	79.1	76.0	91.5	90.6	88.6	88.5

TABLE III

COMPARISONS OF DIFFERENT RESOLUTIONS USING D2REC WITH ViT-B+TRANS ON MVTEC, BTAD, VISA AND BMAD

Tasks	Input Size	MVTEC[1]		BTAD[2]		VisA[3]		BMAD[50]	
Seg	224×224	98.9	74.3	97.6	61.3	99.0	48.5	98.0	60.6
	448×448	98.6	77.6	97.5	65.5	98.6	53.6	97.3	62.0
Cls	224×224	98.9	99.6	96.2	96.6	95.4	96.3	88.6	88.5
	448×448	99.3	99.7	95.5	97.3	97.2	97.9	89.0	88.9

Results on Different Resolutions The results in Tab. III demonstrate that increasing the input resolution to 448×448 generally benefits both anomaly segmentation and classification tasks, with a more significant and consistent positive impact on the pixel-level localization capability. This suggests that the D2Rec benefits from finer spatial details provided by the higher resolution, which is crucial for precise anomaly segmentation.

Qualitative comparisons of anomaly segmentation. In Fig. 4, we visualize the prediction heatmaps of our methods (DRec and D2Rec) and several strong baselines (AE, UniAD, RD and RD*) on MVTEC, BTAD, VisA and BMAD testing images with the unified setting. Our D2Rec provides the most precise predictions of anomaly regions with high confidence, especially on small and slender defects (*e.g.*, contamination on bottle, hole on hazelnut, and crack on title), while the baseline methods exhibit some ambiguous predictions due to the confusion between normal and anomaly samples in feature space. This further suggests that the proposed self-supervised discriminator indeed enhances the discrimination between normal and anomaly samples, and thus alleviates the “weak discrimination” issue in existing reconstruction-based anomaly detection methods.

Comparison of complexity and efficiency Under a unified setting (one model for multiple classes), we compare the number of learnable parameters (M) and inference speed (FPS) of our methods (DRec and D2Rec) with their baselines (AE, MAE and RD) and all competitors, CS-Flow [54],

TABLE IV

COMPLEXITY (#LPs, THE NUMBER OF LEARNABLE PARAMS), EFFICIENCY (FPS) AND PERFORMANCE (P-/I-AUPR AND P-/I-AUROC) COMPARISONS AMONG OUR METHODS (DREC AND D2REC), THEIR BASELINES (AE, MAE, RD AND RD*) AND OTHER COMPETITORS

Metrics	CS-Flow [10]	PaDiM [11]	PatchCore [13]	CFA [14]	DRAEM [15]	SimpleNet [17]	UniAD [21]	MoEAD [24]	OneNIP [47]	AE	MAE Eb4 + Trans	DRec	D2Rec	RD WR50 + Convs	DRec	D2Rec	RD* ViT-B + Trans	DRec	D2Rec
LPs (M) ↓	169.7	950.4	291.3	12.1	97.4	3.9	7.7	4.9	8.4	6.5	6.5	6.5	7.1	83.8	83.8	86.2	56.7	56.7	58.8
FPS ↑	15.4	478.2	18.1	144.2	66.2	100.5	428.7	130.9	220.6	458.4	494.6	402.7	392.5	199.3	135.2	118.7	150.6	106.2	103.9
P-AUPR ↑	33.8	37.8	50.1	37.1	49.6	24.8	44.7	44.7	63.9	22.1	42.2	44.2	61.1	45.9	52.9	67.4	48.0	55.4	74.3
P-AUROC ↑	93.8	95.5	96.4	90.7	85.2	81.0	96.8	97.0	97.9	86.7	95.6	96.3	97.4	94.7	97.0	97.7	97.5	98.2	98.9
I-AUPR ↑	90.2	92.8	96.3	91.0	95.3	90.0	98.9	99.2	99.3	89.6	97.6	97.7	98.9	95.0	97.7	98.6	99.1	99.8	99.6
I-AUROC ↑	81.4	87.5	89.8	80.4	91.4	78.2	96.7	97.7	98.0	77.3	93.5	94.0	96.8	90.9	95.2	97.0	98.7	99.4	98.9

PaDiM [11], PatchCore [13], CFA [14], DRAEM [15], SimpleNet [17], RD [16], UniAD [21], MoEAD [24] and OneNIP [47]. The experimental results are shown in Tab. IV and Fig. 5. The main observations can be summarized as follows:

Firstly, our methods (DRec and D2Rec) and their baselines (AE and MAE) enjoy a reasonable complexity among all competitors if we measure it with the number of learnable parameters. Specifically, our D2Rec (DRec+SSD) with Eb4 +Trans still has fewer parameters compared to the state-of-the-art UniAD [21] (7.1M vs. 7.7M). This benefit mainly comes from two aspects. One is that we remove all components of UniAD, especially layer-wise queries in each decoder layer. The second one is the designed discriminator module as lightweight as possible (only 0.6M).

Secondly, our methods (DRec and D2Rec) with Eb4 +Trans achieve about 400 FPS at inference, which is faster than the popular PatchCore [13] (about 20×) and comparable to state-of-the-arts, such as UniAD [21], MoEAD [24] and OneNIP [47]. We find that D2Rec is faster (392 FPS vs. 131 FPS) than MoEAD in actual inference, despite MoEAD having fewer trainable parameters (4.9M vs. 7.1M). Note that PaDiM [11] also drives a high inference speed (478 FPS) but brings a larger number of parameters (950M).

Thirdly, our final solution (D2Rec) with Eb4 +Trans integrates two proposed proposals, dual-masked reconstruction and self-supervised discriminator, into a framework, which achieves better anomaly segmentation in pixel-level AUPR (61.1%) and anomaly classification in image-level AUROC (96.8%) with fewer parameters (7.1M) and comparable inference speed (393 FPS). Our D2Rec with Eb4 +Trans shows a slight degradation compared to the current state-of-the-art OneNIP [47], but it is worth noting that OneNIP requires one additional normal image at inference.

Last but not least, our D2Rec with ViT-B+Trans still enjoys reasonable inference speed (104 FPS) and achieves the best anomaly detection performance (74.3% in pixel-level AUPR and 98.9% in image-level AUROC) when extended to a transformer-based reversed distillation network with a more powerful backbone, DINOv2 ViT-B/14. This also implies the strong scalability of our method. In other words, the proposed two key components of our method can be freely plugged into reconstructed networks of different architectures and backbones.

C. Ablation Studies

To verify the effectiveness of all proposed components and the effects of hyperparameters, we implement extensive

ablation studies on MVTec with a unified setting with Eb4 +Trans architecture, as shown in Tab. V and Fig. 6.

1) *Dual-Masked Reconstruction*: First, simple random masking is effective for preventing over-fitting. However, simple masked reconstruction only works well with a high masking ratio (90%), while others with low mask ratios still suffer from over-fitting (first row in Fig. 6); Second, the masked reconstruction with low mask ratios causes over-fitting because it may learn a simple copying operation that directly outputs unmasked features as reconstruction. To prevent this simple copying, we present dual-masked reconstruction, DRec, which works well with various mask ratios (second row in Fig. 6). This means that the dual-masked mechanism thoroughly resolves the over-fitting of masked reconstruction. Third, the proposed dual-masked reconstruction is simple yet general, and it can be freely extended to reconstruct networks with different architectures. It is worth noting that most existing works are elaborately designed for specific network architectures.

2) *Self-Supervised Discriminator*: It can be observed that pixel-level AUPR curves (red) are still very low using DRec (second row in Fig. 6) because it is only trained on anomaly-free training data, which may result in weak discrimination between normal and abnormal samples in feature space. To enhance feature discrimination, we propose a self-supervised discriminator to refine reconstruction errors. In the last row of Fig. 6 and Tab. V, it is most effective for anomaly classification and segmentation when integrating dual-masked reconstruction and self-supervised discriminator into a network (*i.e.*, D2Rec). In order to analyze the effects of the self-supervised discriminator, we ablate it from model complexity and synthesis methods. In Tab. Vb, we can see that more convolution blocks and larger convolution dimensions help improve the performance of anomaly detection. In Tab. Vf, increasing the diversity of synthesis methods also contributes to performance improvements. In addition, we empirically find that stopping the gradient in the self-supervised discriminator is helpful to improve performance, as shown in Tab. Va. The stop-gradient effectively isolates the learning of unsupervised reconstruction and self-supervised discriminator, thereby these synthesized anomaly images only train the discrimination. It is worth noting that there is a distribution gap between the synthesized and the real anomaly images. The independent optimization strategy effectively mitigates the overfitting risk of reconstruction networks.

3) *Effects of Hyper-Parameters*: We carefully ablate the effects of other hyperparameters, *e.g.*, the combination coefficient w , the mask ratio, and the weight of the loss function

TABLE V
ABLATION STUDIES ON MVTEC. DEFAULT SETTINGS ARE IN BLUE

(a) Dual-Masked, Discriminator and Stop Gradient							(b) Block and Dim. Number					(c) Coefficient w			
Method	M.R.	D.M.	Dis.	S.G.	I-AUROC	P-AUROC	P-AUPR	#B., #D.	I-AUROC	P-AUROC	P-AUPR	w	I-AUROC	P-AUROC	P-AUPR
AE	0	X	X	X	77.2	86.7	22.1	1, 128	96.5	97.1	58.9	0.10	96.3	97.6	62.7
MAE	50	X	X	X	88.5	76.1	20.5	2, 128	96.8	97.4	61.1	0.25	96.6	97.5	62.4
	90	X	X	X	94.4	95.6	42.2	3, 128	96.7	97.4	61.3	0.50	96.8	97.4	61.1
DRec	50	✓	X	X	94.9	96.2	44.0	2, 64	96.3	97.3	60.4	0.75	96.0	97.1	56.6
D2Rec	50	✓	✓	X	96.5	97.3	60.2	2, 128	96.8	97.4	61.1	0.90	94.9	96.7	49.8
D2Rec	50	✓	✓	✓	96.8	97.4	61.1	2, 256	96.9	97.4	61.5	1.00	94.0	96.3	44.2

(d) Mask Ratio				(e) Loss weight λ				(f) Comparisons of different synthesis methods, and Hybrid is composed of CutPaste and Perlin			
M.R.	I-AUROC	P-AUROC	P-AUPR	λ	I-AUROC	P-AUROC	P-AUPR	Method	I-AUROC	P-AUROC	P-AUPR
10	95.5	97.2	58.1	0.10	96.6	97.3	60.9	Perlin	95.4	96.9	56.8
25	96.5	97.3	60.5	0.25	96.6	97.3	60.9	CutPaste	96.7	97.3	59.3
50	96.8	97.4	61.1	0.50	96.8	97.4	61.1	Hybrid	96.8	97.4	61.1
75	96.6	97.3	61.1	0.75	96.6	97.3	60.9				
90	96.8	97.3	61.0	0.90	96.8	97.3	61.2				

for our D2Rec. The experimental results are reported in Tab. Vc, d, and e. It can be seen that our D2Rec is robust when the mask ratio and loss weights are set to a reasonable interval. Furthermore, it is important to choose a reasonable combination coefficient w for better performance. It will be dependent on the DRec when the coefficient is large, and on the self-supervised discriminator when the coefficient is small.

V. MORE DISCUSSIONS

A. AUROC and AUPR Metrics

For imbalanced anomaly segmentation, the AUROC metric is, in fact, difficult to measure the performance gap for different methods, and the AUPR is a more appropriate metric. The relationship between AUROC and AUPR has been comprehensively studied [51] and some takeaways can be concluded as follows: A random classifier has an AUROC of 50% and an AUPR of the rate of positive examples; An imbalanced dataset might skew ROC-curves and make them look more similar than PR-curves; AUPR is more informative than AUROC for an imbalanced dataset.

Anomaly detection is a typical binary classification with a severe imbalance issue, especially in anomaly segmentation. For example, the proportion of anomaly pixels is only 3.2% on MVTEC testing set, which means that a random classifier only obtains 3.2% AUPR but 50% AUROC for anomaly segmentation. Furthermore, the AUROC (64.0%) is still high (but the AUPR is low, *i.e.*, 4.4%) even if we incorrectly predict all pixels of each anomaly testing image as anomaly ones. Therefore, we believe that AUROC may be suitable to measure image-level anomaly classification and AUPR is better for pixel-level anomaly segmentation because image-level anomaly classification is relatively balanced, while pixel-level anomaly segmentation is extremely imbalanced. This also explains why our method is very significant compared to state-of-the-art when using the AUPR metric and marginal when using the AUROC metric. In addition, the reliability of AUPR metric as a measure is also corroborated in qualitative comparisons in Fig. 4.

B. Comparisons of Separated and Unified Models

Most existing anomaly detection methods mainly focus on training separate models for different classes. However,

TABLE VI
COMPARISONS OF SEPARATE AND UNIFIED MODELS. THE SUPERScript * MEANS THAT THE RESULTS ARE RE-PRODUCED BY US

Methods	Setting	I-AUROC	P-AUROC	P-AUPR
UniAD*	separate	94.4	96.3	44.5
D2Rec (Ours)	separate	96.5	97.5	60.4
D2Rec (Ours)	unified	96.8	97.4	61.1

this separate paradigm may not be practical in real-world applications, as it may require high memory consumption and storage overhead, especially when increasing the number of categories. In addition, these separate models only perform well on simple distributions (*i.e.*, one model for one class) but fail to handle such a challenging task (one model for multiple classes). Therefore, the unified paradigm is both more practical and more challenging. Furthermore, it is necessary to study unified anomaly detection from a foundational model perspective.

Following previous separate arts, we train separate models, which are learned in the same way as our unified model, but on simple data distributions (one model for one class). And we compare separate models with unified ones in Table VI. For fair comparisons, we re-implement UniAD for separate training because the original paper doesn't report the AUPR results. *Our D2Rec is better than the re-produced UniAD in both image-level classification and pixel-level segmentation with AUROC, and significantly better in pixel-level segmentation with AUPR (61.1% vs. 44.5%). More importantly, there is no significant performance drop from the separate case to the unified case (60.4% vs. 61.1%).* In contrast, most existing separate (*i.e.*, one model for one class) methods suffer from a significant performance drop when they are extended to complex and challenging distributions (*i.e.*, one model for multiple classes).

C. Unsupervised Reconstruction on Real Normal Images and Self-Supervised Discriminator on Pseudo Anomaly Images

We learn the dual-masked reconstruction model by optimizing the reconstruction loss between the reconstructed and original features of normal training images. This process allows the model to learn how to detect anomalies because

TABLE VII

COMPARISONS OF RECONSTRUCTION- AND MEMORY-BANK-BASED MODELS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Category	#small	#large	#large/#all (%)	PatchCore	UniAD	D2Rec
Tile	4	80	95.2	59.9	44.2	77.1
Bottle	5	58	92.1	72.1	68.0	83.6
Cable	14	78	84.8	56.0	53.1	53.3
Metal Nut	16	77	82.8	80.8	49.5	68.4
Wood	14	46	76.7	52.1	37.8	69.5
Pill	98	43	30.5	59.0	40.7	31.7
Grid	48	9	15.8	12.0	22.2	43.7
Leather	78	14	15.2	41.3	33.7	70.2
Capsule	93	16	14.7	36.7	46.5	48.3
Screw	119	0	0.0	8.9	24.5	35.2

anomaly images are absent in normal training samples, and they should not be accurately reconstructed. To enhance the model’s discrimination ability, it is crucial to introduce pseudo (synthesized) anomaly images for self-supervised training. Notably, the pseudo anomaly images are generated by applying region-level perturbations or disruptions to freely available normal images, resulting in features that naturally contain both positive (normal) and negative (anomaly) samples. Therefore, it is possible to train the self-supervised discriminator using Eq. 8. The self-supervised discriminator refines reconstruction errors from low to high resolution for accurate anomaly segmentation. It is worth noting that it is effective to independently train the reconstruction network with real normal images and the discrimination network with pseudo anomaly images due to the distribution gap between real and synthesized anomaly images. Here, the independent training is effectively achieved by applying a stop-gradient operation on the discrimination network, ensuring that the two networks are trained separately yet effectively.

D. Comparisons of Reconstruction and Memory-Bank Methods

Reconstruction-based methods are vulnerable to large-size defects compared to memory-bank methods. However, our D2Rec exhibits excellent performance on both large and small defects. We roughly define large defects as more than 800 pixels with 224×224 resolution, and others as small defects. We list top5 and bottom5 classes on MVTec and the corresponding pixel AUPR results of PatchCore [13], UniAD [21], and our D2Rec in Table VII. We can see that reconstruction-based UniAD is good at detecting small defects (bottom5) and memory-bank-based PatchCore is better at detecting large defects in many cases. However, this issue can be effectively balanced using our D2Rec method, which significantly outperforms PatchCore (4 classes in top5) and UniAD (4 classes in bottom5).

VI. CONCLUSION

In this paper, we rethink reconstruction networks for unified unsupervised vision anomaly detection, and then find that they severely suffer from over-fitting and result in unsatisfactory performance because of “identity shortcut” and “weak discrimination”. To address these two issues, we propose an embarrassingly simple but effective D2Rec, which is mainly composed of a dual-masked reconstruction and self-supervised

discriminator. The dual-masked reconstruction is simple and general, and it provides essential insights into addressing “identity shortcut” that all masked positions are reconstructed using unmasked features implemented with a simple dual-masked mechanism. Meanwhile, it is robust for mask ratio and doesn’t involve other additional hyperparameters and thus greatly simplifies state-of-the-art UniAD. Furthermore, when integrating the dual-masked reconstruction and self-supervised discriminator into a unified network, D2Rec achieves competitive performance on image-level anomaly classification and pixel-level anomaly segmentation. Furthermore, the proposed dual-masked reconstruction and self-supervised discriminator can be easily used as a general plug-in into various reconstruction networks, consistently improving their performance.

A. Limitation

Our method requires a doubling of computational cost during training due to the self-supervised discriminator on additional synthesized images. However, the computational consumption is comparable to its competitors at inference because the synthesized branch will be removed.

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